

A Comprehensive Picture of the Structures, Energies, and Bonding in the Alanine Dimers

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High level quantum mechanical computations and extensive stochastic searches of the potential energy surfaces of the Alanine dimers uncover rich and complex structural and interaction landscapes. A total of 416 strongly bound (up to 13.4 kcal mol⁻¹ binding energies at the DLPNO-CCSD(T)/6-311++G(d,p) level corrected by the basis set superposition error and by the zero point vibrational energies over B3LYP-D3 geometries), close energy equilibrium structures were located, bonded via 32 specific types of intermolecular contacts including Y...H-X primary and Y...H-C secondary hydrogen

bonds, H...H dihydrogen contacts, and non conventional anti-electrostatic Y^{δ-}...X^{δ-} interactions. The putative global minimum is triply degenerate, corresponding to the structure of the common dimer of a carboxylic acid. All quantum descriptors of chemical bonding point to a multitude of weak individual interactions within each dimer, whose cumulative effect results in large binding energies and in an attractive fluxional wall of non-covalent interactions in the interstitial region between the monomers.

1. Introduction

Protein chemistry is one of the most important topics in molecular biochemistry. The primary structures of these large molecules comprise sequences of a set of only 20 amino acids of general formula R-NH₂CO₂H.^[1] Proteins are involved in virtually every aspect of cell and virus functions, from their inner workings to the interactions with the environment. More generally, it may be argued that proteins are involved in critical life related processes at the molecular level, there is even a protein interaction world hypothesis for the origin of life. The very existence of individual, specialized proteins, is an extraordinary occurrence: consider that a total of 20! ≈ 2.4 × 10¹⁸ distinct primary sequences, each one with its own rich and complex conformational landscape and secondary and tertiary structures, are possible for a very simple (in the protein world) molecule containing just one of each of the 20 amino acids.

Non-covalent contacts determine every aspect of protein-protein interactions and thus, because of the large number of possible binary contacts, a systematic, thorough knowledge of how proteins recognize and bind to each other is an utopic goal. Since primary and secondary hydrogen bonds are among

the most important factors determining protein-protein interactions,^[2] the number of possible residue to residue contacts when a pair of proteins are interacting is staggering, thus, the number of structural possibilities for a given protein dimer is limited but extremely large. Let us illustrate this point by progressively increasing the complexity of the system. First, consider Glycine, the smallest amino acid with formula C₂H₅NO₂.^[3] There are five hydrogen atoms and three electro-negative atoms, thus, there are a total of 5 × 3 = 15 possibilities of hydrogen bonding for the Glycine dimers under the classical X-H...Y hydrogen bond perspective. Now consider insulin, one of the smallest proteins, containing just 51 amino acids with formula C₂₅₇H₃₈₃N₆₅S₆O₇₇.^[4,5] Each insulin has 383 protons and 148 proton acceptor sites, thus, there are at least 383 × 148 = 56684 ≈ 6 × 10⁴ possibilities for hydrogen bonding in isolated insulin dimers just in the classical donor-acceptor direction. The complexity of the situation grows exponentially with the size of the protein, thus, since commonly occurring proteins contain in excess of 1000 amino acids and may also exhibit salt bridges, π→π, and hydrophobic contacts, the number of binary interacting possibilities is simply too large to be properly accounted, not to mention that multiple contacts are also possible within a given pair of amino acids.^[2]

The above argumentation of combinatorial possibilities for primary sequences of proteins and for residue to residue interactions in protein dimers is not purely academic, quite to the contrary, because of the immense number of realizable possibilities leading to increasing entropy, it is intimately linked to the crucial problem of life, its origins, evolution, and genetic diversity, via the second law of thermodynamics. Let us take virus evolution as an example of an entropy driven process.^[6] The process of how viruses evolve to survive and thrive in local environments is well understood in terms of the molecular changes induced by the evolutive pressure in a given environment. For most viruses, the probability of individual mutations in a given primary sequence is of the order of 10⁻⁶, however,

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because of the large number of reproductive events and short reproductive cycles, each individual mutation eventually occurs no matter how small the probability, and when a particular change helps the virus adapt to the local environment, the new variants become the dominant forms. Three typical examples are the HIV virus, which as summarized by Rabaut and coworkers^[7] has a 3×10^{-5} mutation rate per base and generation that helps it overcome the immune system, the Influenza virus, which mutates so rapidly that annual reformulations of the vaccines are needed to fight new variants,^[8] and the SARS-COV-2 virus, for which new variants are reported at a seemingly faster pace as the pandemic goes on, we have for example seen the emergence of the $\alpha, \beta, \gamma, \delta$, etc., dominant variants, with their magnified infectious power brought about by simple individual mutations in the spike protein that lead to higher affinities towards the ACE2 receptor, favoring (from the virus stand point) the initial virus–cell molecular recognition and attachment, events that trigger the infectious cycle.

Given the sophisticated understanding of protein structure in present days, it is unfortunate that the fundamental reasons behind protein–protein binding are for the most part collectively swept under the too general categorization of hydrogen bonds, salt bridges, $\pi \rightarrow \pi$, and hydrophobic interactions, without much scrutiny of specific individual interactions. This is quite worrisome because the very concept of intermolecular interaction is too broadly treated where it seems to matter most. Indeed, when analyzed under the quantum mechanics formalism, hydrogen bonds for example reveal a picture of donor $\rightarrow$ acceptor ($d \rightarrow a$) orbital interaction^[9–12] which transcends the primitive $X^{\delta-} \cdots H^{\delta+} \cdots Y^{\delta-}$ electrostatic vision that is a common place in general chemistry and biology textbooks.

In this work, we chose Alanine, the smallest amino acid with an α chiral carbon and the amino acid with the second smallest side chain (second only to Glycine) as a test case, and attempt to provide an in-depth, systematic analysis of all the possibilities for individual ALA–ALA contacts, focusing on the nature and strength of the interactions. To this end, we comprehensively explore the configurational space of the dimers via a stochastic search algorithm, calculate highly correlated interaction energies, and dissect the nature of bonding interactions.

1.1. Methods

Sampling the configurational space of the ALA–ALA dimers is a demanding task: First, there are at least 21 distinct hydrogen bonding possibilities within each dimer (7 protons and 3 electronegative atoms per amino acid). Second, there is the possibility of other binary contacts (dihydrogen bonds, etc.). Third, multiple contacts between the two monomers are not uncommon. Fourth, there is the structural complexity of the monomers, which has been extensively discussed in several works^[13–15] and we summarize next. Two Alanine conformations of the L stereoisomer, estimated to be separated by ≈ 0.5 kcal mol^{−1}, have been experimentally detected in the gas phase via several spectroscopies^[15–18] (the zwitterionic form has not been detected in the gas phase^[19]), however, there are probably many more structures escaping the detection limits. Indeed, just for the L configuration, as many as 13 equilibrium structures in small energy windows have been calculated.^[13–15] A number of factors contribute to this structural variety: there are the possibilities of *syn*, *anti*, and *gauche* conformations about the C–C bond in the O=C–C–N fragment (strictly speaking, most structures fall into the quasi-*anti*, quasi-*syn*, and quasi-*gauche* categories), there are small rotational barriers for the O–H bond and for the NH₂ and methyl groups, and there is the possibility of weak intramolecular hydrogen bonds. In addition, each monomer conformation has the possibility of the two L, D stereoisomers. As a representative sample of the monomers, Figure 1 shows the five lowest energy isomers of Alanine according to the gas phase free energies at room conditions.

The large number of calculated equilibrium geometries for the Alanine monomer poses a serious problem for practical calculations of the dimers: there a total of 13 L and 13 D structures,^[13] thus, a simple combinatorial analysis shows that strictly speaking, there are $26!/(24!2!) + 26 = 351$ distinct ALA–ALA potential energy surfaces to explore. However, we argue that since all monomers fall within a small energy range, and since upon formation of the dimers most of the weak intramolecular interactions and rotational barriers within the monomers either cease to exist or are not comparable to the energies stabilizing the dimers (for example, if the O–H bond in ALA3, Figure 1 is to donate its proton to an intermolecular hydrogen bond, the intramolecular O–H–N weak interaction ceases to

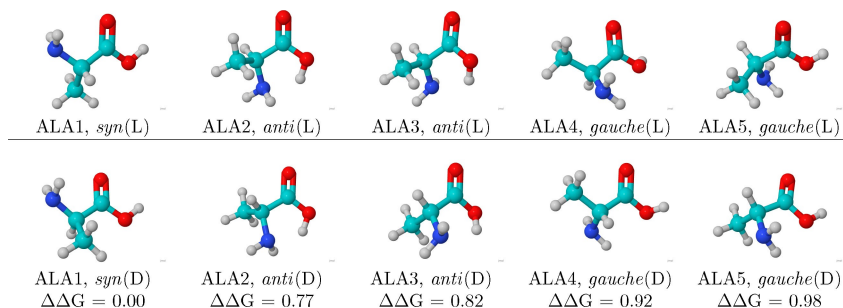


Figure 1. Lowest energy isomers of Alanine. ALA1(L) and ALA2(L) have been experimentally detected in the gas phase.^[13–15] The *anti*, *syn*, *gauche* notation is used to establish the positioning of the amino and carbonyl groups with respect to the C–C bond in the O=C–C–N sequence.

exist in the dimer), a reduced sampling of possibilities will afford a statistically sound picture of the entire ALA...ALA interaction problem, in other words, rigorously exploring the 351 PESs would lead to lots of redundant information. Accordingly, our pragmatic approach to sample the structural space of Alanine dimers may be summarized as follows.

We chose both L, D stereoisomers of ALA1, ALA4 (Figure 1). The corresponding potential energy surface for the intermolecular interactions leading to stable minima for each one of the 10 ALA...ALA pairs that can be constructed from these four structures were explored via random walks under Simulated Annealing conditions using the ASCEC algorithm.^[20–22] Starting with ALA1(L), the relaxed scan in Figure S1 and the energy differences in Table S1 show that dispersion and the size of the basis set play crucial roles not only on the energetics but also on the molecular geometries of the monomers, with B3LYP-D3 and MP2 affording minimal differences, thus, we report final equilibrium structures for the dimers using the dispersion corrected B3LYP-D3 functional^[23–27] in conjunction with the 6–311++G(d,p) basis set. All geometry optimizations and characterizations as true minima via harmonic vibrational analysis were carried out using the Gaussian16 suite of programs.^[28] Highly correlated DLPNO-CCSD(T)/6–311++G(d,p) energies using the TightPNO truncation threshold were obtained using the ORCA package^[29–31] for all B3LYP-D3 optimized dimers. The TightPNO truncation setup has been suggested for accurate prediction of the binding energies of non-Covalent bonded clusters.^[32–34]

Once the structural problem is solved via the above random search of the potential energy surfaces, we proceeded to analyze the intermolecular interactions that keep each dimer as a stable unit using the tools provided by the Quantum Theory of Atoms in Molecules (QTAIM),^[35] by the natural bond orbitals (NBO)^[11,36] and by the non-covalent interactions (NCI).^[37,38] These are well established tools to dissect bonding interactions, thus, we make no attempt to discuss their merits and flaws, the interested reader is advised to consult the vast specialized literature devoted to both methods and its applications.^[10,39–42] In short, we construct a picture of the bonding situation by analyzing the properties of the bond critical points (BCPs, r_c) arising from intermolecular contacts, and by pinpointing and analyzing the involved molecular NBOs from each fragment. Specific descriptors include the electron density $\rho(r_c)$, its Laplacian $\nabla^2\rho(r_c)$, the total, kinetic, and potential energy densities $\mathcal{H}(r_c) = \mathcal{G}(r_c) + \mathcal{V}(r_c)$, the virial ratio $|\mathcal{V}(r_c)|/\mathcal{G}(r_c)$, and the second order perturbation interaction energy between donor and acceptor orbitals $E_{d \rightarrow a}^{(2)}$. All QTAIM descriptors were obtained using the AIMall suite.^[35] NBO7,^[36] as implemented in Gaussian16, was used for the NBO calculations. The NCIPLOT program^[43] was used to derive the non-covalent interaction surfaces.

2. Results and Discussion

2.1. Geometries and Energies

A staggering structural diversity for the Alanine dimers is uncovered after stochastic searches of the potential energy surfaces for each one of the 10 possible monomer...monomer pairs resulting from combinations of the four ALA1(L), ALA1(D), ALA4(L), ALA4(D) structures (Figure 1). A total of 416 well defined equilibrium structures stabilized by the 32 different types of intermolecular interactions listed in Table 1 and schematically shown in Figure 2 with the corresponding donor and acceptor NBOs, were found. The B3LYP-D3/6–311++G(d,p) potential energy surface for the Alanine dimers contains three dominant isomers (>30% population each) with the same geometries but different stereochemistry, thus, the well known structure for the carboxylic acid dimers (structure 1 in Figure 2) has three equivalent forms: L...L, L...D, D...D. Notice that contrary to the putative global minima, in most other dimers, a simple inversion of the configuration about one or both monomers does not lead to equivalent structures and may not even lead to equilibrium geometries because the chiral fragments may be involved in intermolecular interactions. We use a simple notation to address the dimers based on the Gibbs energies relative to the lowest energy structure. In this notation, D₁, D₂, and D₃ are the three-fold degenerate putative global minima ($\Delta\Delta G=0$), and so on. Cartesian coordinates and energetic data (Table S2) for the entire set of 416 dimers are provided in the Supplementary Material.

The large number of equilibrium structures within small energy windows obtained from our limited sampling (only 10 out the total 351 possible PESs were explored, yet we found 416 structures within $\Delta\Delta G \approx 16 \text{ kcal mol}^{-1}$ of the lowest energy structure), and the large number of distinct types of intermolecular contacts for the dimers of Alanine, are certainly overwhelming. Indeed, combinations of C₃H₇NO₂ monomers into dimers should lead to $7 \times 3 = 21$ possible intermolecular contacts from the classical hydrogen bond picture, which might further be reduced into the first 1–6 distinct primary and 7–12 secondary hydrogen bonds listed in Table 1, however, the structural richness is considerably higher. A summary of all intermolecular interactions found in this work is listed in Table 1 and illustrative samples are provided in Figure 2. In this context, the C=O...H–O interaction, responsible for the dimerization of carboxylic acids, is not the only stabilization, although it is present in the dimers that account for $\approx 98\%$ of the population (Table S2). One of the most important results from this study is that we are able to rationalize the large structural diversity of the Alanine dimers and by extension the seemingly insurmountable structural diversity of protein...protein interactions leading to the astonishing variety of life, by invoking not only traditional primary and secondary hydrogen bonds, but other types of intermolecular contacts, including H...H dihydrogen interactions,^[44,45] O...C, O...O, O...N, N...C, N...N, and C...H contacts.

There are studies dealing with the structures and energies of Alanine dimers.^[46] A remarkable observation drawn from Table 2 is that Alanine dimers are strongly bound: the DLPNO-

Table 1. Properties of the 32 types of intermolecular interactions stabilizing the Alanine dimers (Figure 2). Only the largest $-E_{d \rightarrow a}^{(2)}$ energy in kcal mol $^{-1}$ for each type is listed along with the corresponding individual dimer containing it.

Label	Type	Number of occurrences	Orbitals involved	$-E_{d \rightarrow a}^{(2)}$	Dimer
Primary hydrogen bonds					
1	C=O ... H-O	100	$n_O \rightarrow \sigma_{H-O}^*$	21.50	D ₁
2	C=O ... H-N	207	$n_O \rightarrow \sigma_{H-N}^*$	3.88	D ₉₈
3	H-O ... H-O	48	$n_O \rightarrow \sigma_{H-O}^*$	10.94	D ₁₅₇
4	H-O ... H-N	47	$n_O \rightarrow \sigma_{H-N}^*$	1.74	D ₄₀₈
5	N ... H-O	78	$n_N \rightarrow \sigma_{H-O}^*$	35.68	D ₂₄
6	N ... H-N	85	$n_N \rightarrow \sigma_{H-N}^*$	9.63	D ₂₉₀
Secondary hydrogen bonds					
7	C=O ... H-C $_{\alpha}$	141	$n_O \rightarrow \sigma_{H-C}^*$	1.01	D ₂₆₅
8	C=O ... H-C $_{\beta}$	203	$n_O \rightarrow \sigma_{H-C}^*$	0.99	D ₄₁₃
9	H-O ... H-C $_{\alpha}$	37	$n_O \rightarrow \sigma_{H-C}^*$	1.31	D ₄₀₉
10	H-O ... H-C $_{\beta}$	73	$n_O \rightarrow \sigma_{H-C}^*$	0.96	D ₃₉₉
11	N ... H-C $_{\alpha}$	67	$n_N \rightarrow \sigma_{H-C}^*$	3.69	D ₁₄₁
12	N ... H-C $_{\beta}$	96	$n_N \rightarrow \sigma_{H-C}^*$	2.47	D ₄₀₆
Dihydrogen contacts					
13	O-H ... H-C $_{\beta}$	7	$\sigma_{C-H} \rightarrow \sigma_{H-O}^*$	0.46	D ₃₄₄
14	N-H ... H-N	2	$\sigma_{N-H} \rightarrow \sigma_{H-N}^*$	0.06	D ₁₅₀
15	N-H ... H-C $_{\alpha}$	11	$\sigma_{N-H} \rightarrow \sigma_{H-C}^*$	0.23	D ₃₅₂
16	N-H ... H-C $_{\beta}$	19	$\sigma_{N-H} \rightarrow \sigma_{H-C}^*$	0.18	D ₃₄₇
17	C $_{\alpha}$ -H ... H-C $_{\alpha}$	22	$\sigma_{C-H} \rightarrow \sigma_{H-C}^*$	0.34	D ₃₀₆
18	C $_{\beta}$ -H ... H-C $_{\beta}$	42	$\sigma_{C-H} \rightarrow \sigma_{H-C}^*$	0.29	D ₁₆₅
19	C $_{\alpha}$ -H ... H-C $_{\beta}$	77	$\sigma_{C-H} \rightarrow \sigma_{H-C}^*$	0.29	D ₃₅₇
O ... C contacts					
20	C=O ... C=O	34	$n_O \rightarrow \pi_{C=O}^*$	1.30	D ₄₁
21	C=O ... C $_{\alpha}$ -C	8	$n_O \rightarrow \sigma_{C-C}^*$	0.26	D ₁₇₈
22	H-C $_{\beta}$... C=O	4	$\sigma_{H-C} \rightarrow \pi_{C=O}^*$	0.17	D ₄₀₀
23	H-O ... C=O	17	$n_O \rightarrow \pi_{C=O}^*$	0.83	D ₉₁
O ... O contacts					
24	C=O ... O=C	25	$\pi_{C=O} \rightarrow \pi_{O=C}^*$	0.61	D ₃₂₉
25	C=O ... O-H	19	$n_O \rightarrow \pi_{O=C}^*$	0.72	D ₂₆₈
26	H-O ... O-H	19	$n_O \rightarrow \sigma_{O-H}^*$	0.43	D ₃₆₉
O ... N contacts					
27	C=O ... N	12	$n_N \rightarrow \pi_{O=C}^*$	0.99	D ₂₃₇
28	H-O ... N-H	8	$n_N \rightarrow \sigma_{O-H}^*$	0.23	D ₁₀₆
N ... C contacts					
29	N ... C=O	14	$n_N \rightarrow \pi_{C=O}^*$	1.66	D ₁₈₃
30	H-N ... C $_{\beta}$ -H	3	$\sigma_{H-N} \rightarrow \sigma_{C-H}^*$	0.09	D ₂₄₅
N ... N contacts					
31	N ... N-H	3	$n_N \rightarrow \sigma_{N-H}^*$	0.19	D ₁₃₇
C ... H contacts					
32	O=C ... H-C $_{\alpha}$	50	$\sigma_{C-H} \rightarrow \pi_{C=O}^*$	0.40	D ₃₈₆

CCSDT binding energy with B3LYP-D3 ZPE corrections for the degenerate global minimum is 17.2 kcal mol $^{-1}$. For comparison, the water dimer, the archetypal hydrogen bonded system is reported to be stabilized by 5.5 kcal mol $^{-1}$ with respect to the isolated monomers at the same DLNPO-CCSD(T) level of theory

and the CCSD(T) binding energies of the carbonic and formic acid dimers are 19.0 and 12.83 kcal mol $^{-1}$, respectively.^[47,48] Accurate treatment of electron correlation is of crucial importance to calculate binding energies:^[46] reported 14.61 kcal mol $^{-1}$ at the DLNPO-CCSD(T)/def2-TZVP//B3LYP/6-311++G(d,p) for

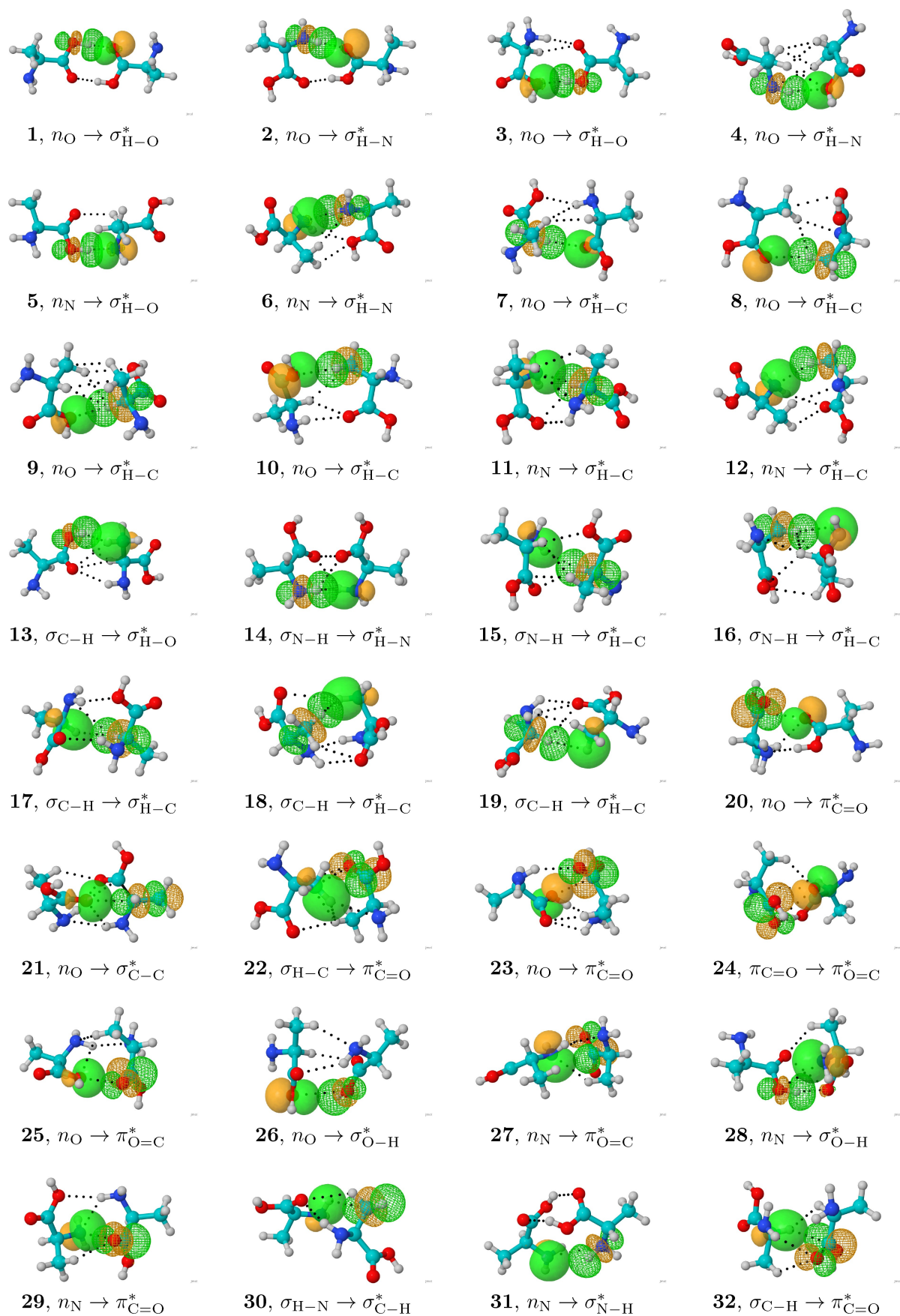


Figure 2. Structural and orbital representation within the NBO picture of the 32 intermolecular contacts stabilizing the Alanine dimers.

Table 2. Binding energies in kcal mol⁻¹ for a representative sample of Alanine dimers. See Table S2 in the Supplementary Material for the complete list of 416 dimers. $\Delta\Delta G$: Gibbs energies relative to the global minimum. We list electronic ($E_{B3LYP-D3}$), DLPNO-CCSD(T) with TightPNO truncation setup (E_{CC}), and DLPNO-CCSD(T) corrected for the basis set superposition error by the counterpoise method of Boys and Bernardi^[49] ($-BE_{CC,BSSE}$) binding energies. All energies corrected by the B3LYP-D3 vibrational ZPE. n_c : number of intermolecular contacts. See Table 1 and Figure 2 for the specific contact types.

Dimer	Stereochemistry	$\Delta\Delta G$	$\Delta E_{B3LYP-D3}$	ΔE_{CC}	$-BE_{B3LYP-D3}$	$-BE_{CC}$	$-BE_{CC,BSSE}$	n_c	Contact type
D ₁	L...L	0.0	0.0	0.0	20.9	17.2	13.4	2	1
D ₂	D...L	0.0	0.0	0.0	20.9	17.2	13.4	2	1
D ₃	D...D	0.0	0.0	0.0	20.9	17.2	13.4	2	1
D ₄	L...L	1.3	1.5	1.3	19.4	15.9	12.1	2	1
D ₁₈	L...D	3.8	4.8	4.2	16.1	13.0	9.4	2	5, 7
D ₂₅	L...L	4.4	5.0	4.4	15.9	12.8	9.3	3	5, 7, 8
D ₂₀	L...D	4.4	2.4	2.7	19.5	15.1	9.2	6	2, 5, 23
D ₂₃	L...D	4.4	5.2	4.9	15.7	12.3	8.6	3	2, 5, 8
D ₂₇	D...D	5.6	3.5	3.5	17.9	14.0	7.4	5	5, 8, 23
D ₂₈	D...D	5.6	6.7	5.7	14.2	11.5	8.0	2	2, 5
D ₃₃	L...L	5.6	7.1	6.1	13.8	11.1	7.3	2	5, 8
D ₄₇	L...L	6.3	6.1	4.8	14.8	12.4	7.6	3	5, 8, 20
D ₆₀	L...D	6.9	7.4	5.7	14.0	11.8	7.1	2	1, 5
D ₁₀₅	D...D	9.4	9.2	7.6	12.2	9.9	5.6	4	1, 6, 20, 32
D ₃₁₈	L...D	14.4	15.6	12.8	6.3	5.1	1.5	8	7, 8, 10, 24
D ₃₇₇	L...L	15.1	16.0	13.6	5.9	4.2	0.8	7	4, 6, 10, 11, 12, 19, 32

their global minimum; pure B3LYP-D3/6-311++G(*d,p*) binding energies calculated here sensibly depart from that number at 20.9 kcal mol⁻¹ for the degenerate global minimum, and our own DLPNO-CCSD(T)//B3LYP-D3 binding energies are 17.2 kcal mol⁻¹ using the 6-311++G(*d,p*) basis set, however, accounting for the basis set superposition error via the counterpoise corrections of Boys and Bernardi^[49] yields 13.4 kcal mol⁻¹, which is substantially closer to the values reported by and to the experimental binding energies of common carboxylic acid dimers, i.e., to the 14.2 kcal mol⁻¹ reported for the dissociation of the formic acid dimer,^[50] which exhibits the exact same stabilizing pattern as the global minimum of the alanine dimers. The collective results exposed in Table 2 and Table S2 serve as a cautionary note on the need to accurately treat both, correlation and basis set superposition error effects if meaningful interaction energies are to be obtained. Another very interesting observation is that the number of intermolecular contacts is not directly related to the binding energy as seen for example in D₃₁₈, which contains 8 intermolecular individual interactions between the two monomers but is 11.9 kcal mol⁻¹ below D₁, a structure with two contacts, in the DLPNO-CCSD(T) energy scale with ZPE and BSSE corrections (Table 2). This *a priori* counter intuitive finding is rationalized by realizing that after the main interaction is in place, some other interactions with smaller stabilization power appear as a consequence of the geometry, thus, the interactions may be conveniently divided into structure determining vs consequence of the structure.

The diversity of intermolecular contacts in the Alanine dimers is exposed with the help of the radial distributions plotted on the top left panel of Figure 3. Taking as reference the dashed vertical line at ≈ 1.97 Å which marks the center of the radial distribution for intermolecular interactions in small water clusters,^[21,51–54] it is interesting to see that there is a large number of really short and really long HBs. Notice that the

radial distribution of the strong C=O...H-O contacts (interaction 1) peaks at ≈ 1.74 Å, with individual cases close to 1.5 Å. At the same time, the radial distribution for the very weak N...H-C _{β} contacts (interaction 12) peaks at ≈ 2.88 Å with a few cases longer than 3.0 Å. For the set of 416 structures reported in this work, the most frequent types of interactions, as labeled in Figure 2 and in Table 1, are, 207 C=O...H-N primary hydrogen bonds (interaction 2), 203 C=O...H-C _{β} secondary hydrogen bonds (interaction 8), 180 dihydrogen contacts (interactions 13–19), 141 secondary C=O...H-C _{α} hydrogen bonds (interaction 7), 100 C=O...H-O primary hydrogen bonds (interaction 1) and 78 N...H-O primary HBs (interaction 5). Broadly speaking, there seems to be an inverse correlation between interaction strength and the width of the band (see below for the quantification of interaction strengths).

2.2. Dissection of Bonding Interactions

It is worth noticing that Weinhold^[55] has shown an equivalency between descriptors of chemical bonding derived from Natural Bond Critical Points (NBCPs) and QTAIM, notwithstanding, bonding interactions are described here using the results of NBO and traditional QTAIM methods.

We identified a total of 32 distinct types of intermolecular interactions, listed in Table 1, as being responsible for the stabilization of the vast number of dimers located in this work. Besides the structure determining vs structural consequences, a general division into four sets of interactions with common features may be offered: primary hydrogen bonds, secondary hydrogen bonds, dihydrogen interactions and non conventional or exotic contacts. We provide an in depth description of these stabilizing interactions using the tools provided by QTAIM^[40,41,56] and NBO^[9–12] plotted in the various panels in Figure 3. A scheme of the donor→acceptor orbital interaction in the NBO picture

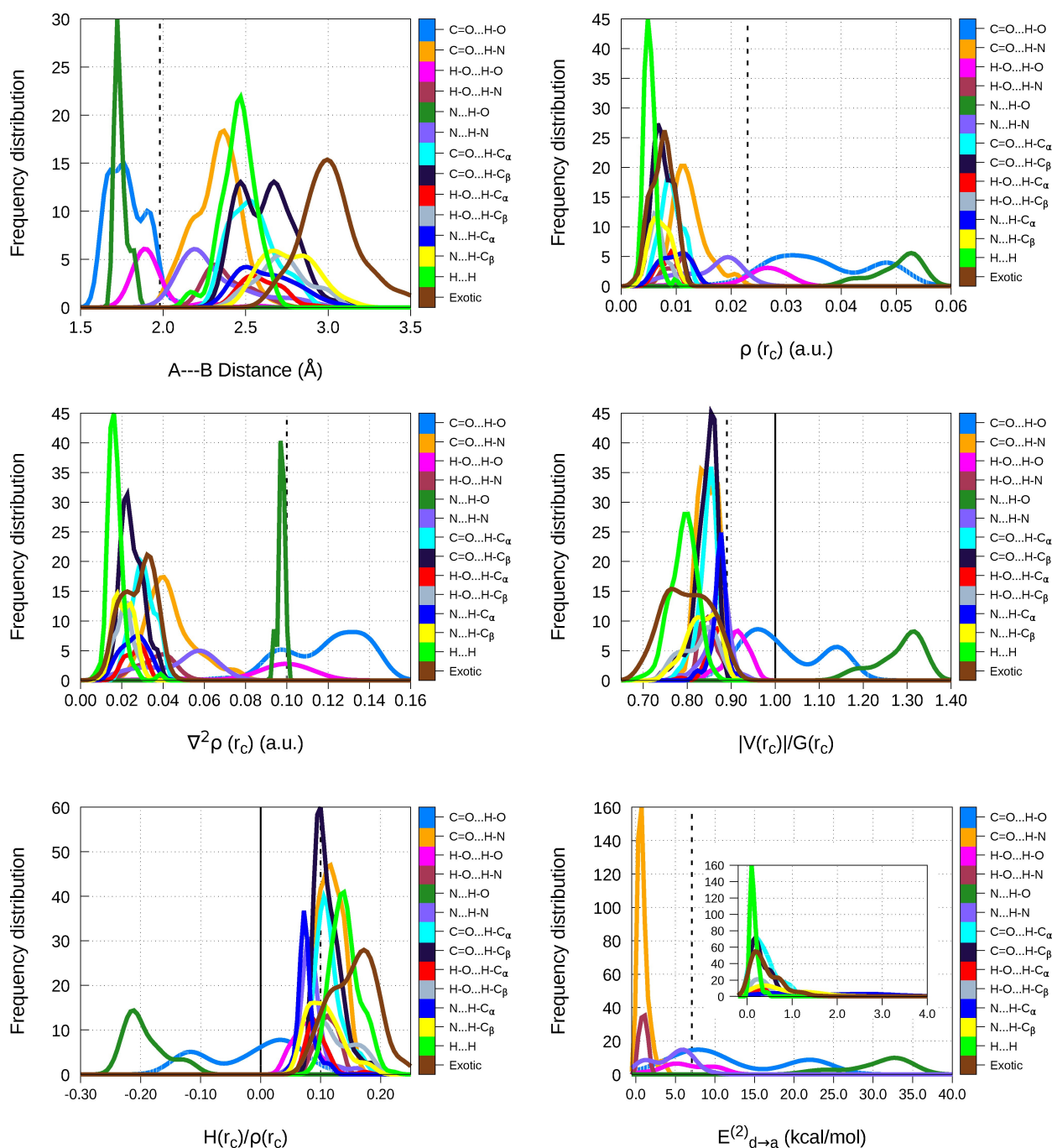


Figure 3. Descriptors of the 32 bonding interactions shown in Figure 2 and listed in Table 1. In all cases, dotted vertical lines mark the properties of the reference water dimer and solid vertical lines establish QTAIM boundaries separating interactions by type (see text).

involved in each case is provided in Figure 2. NCI surfaces are plotted in Figure 4. Without a single exception, all interactions are characterized by positive Laplacians of the electron densities at bond critical points, effectively describing local depletion of charge, associated to long range, weak interactions. Nonetheless, a few cases have negative total energy densities and virial ratios larger than 1 and should thus be considered as of intermediate strength.^[56] As a general rule, as expected, interaction distances are inversely correlated with

accumulations of electron densities at the bond critical points (the peaks to the right of the A---B distances plot shift to the left on the $\rho(r_c)$ plot) and are directly correlated with interaction energies. This result is fully consistent with well documented exponential decays of the electron density as a function of atom separation.^[51,57–59] We analyze next the nature of bonding interactions within each set.

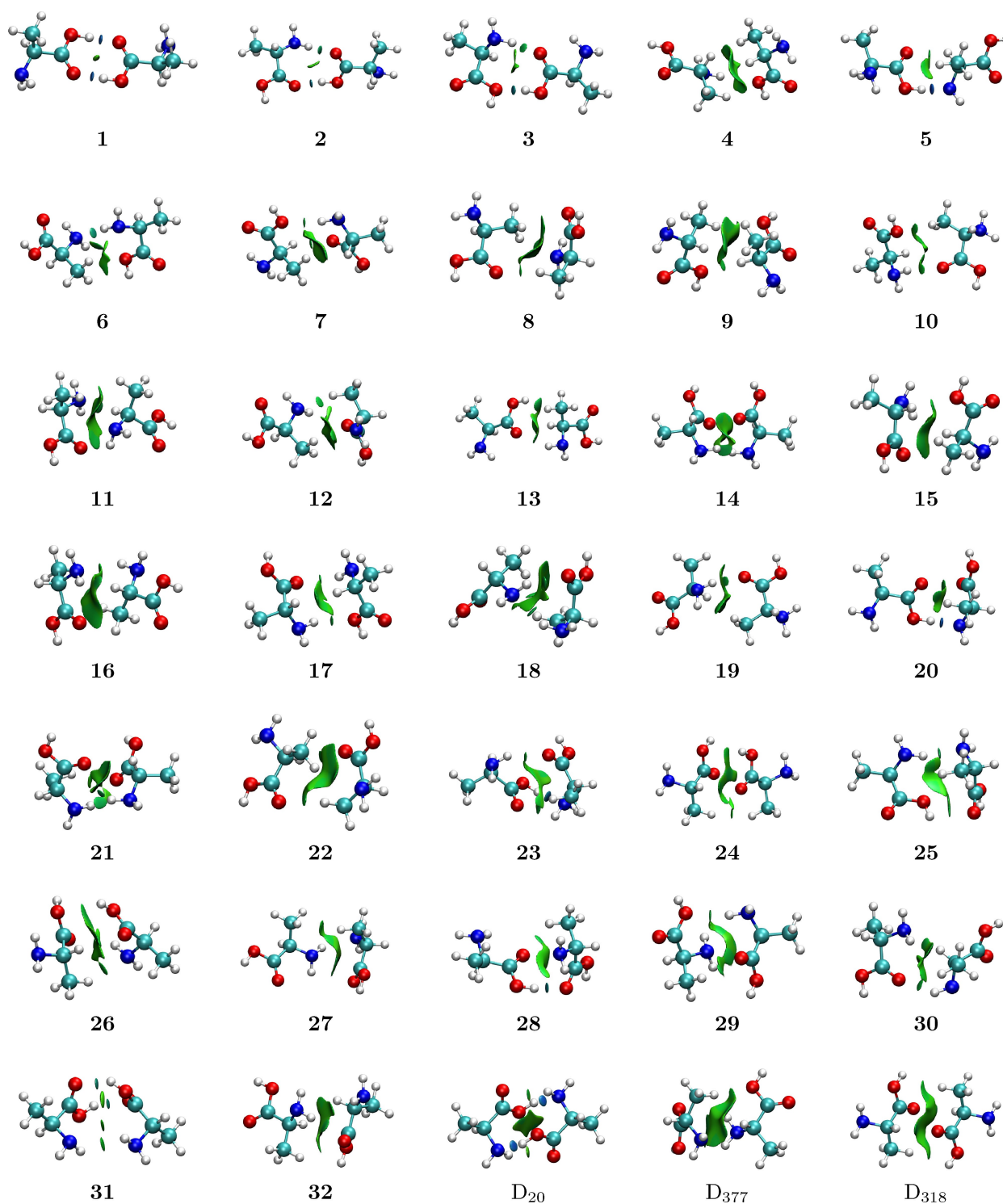


Figure 4. Fluxional walls of attractive (green surfaces according to the NCIPLOT^[43] color code) non-covalent interactions stabilizing the Alanine dimers. In addition to the 32 structures of Figure 2, we include the three dimers with the largest number of intermolecular contacts.

2.2.1. Primary Hydrogen Bonds

Interactions 1–6 in Table 1 and in Figure 2. These are by far the strongest interactions (Figure 3) and are the structure determining factors for most dimers. As a general rule (with a few

exceptions in the H–O...H–O case), in the $\phi_d \rightarrow \phi_a$ scheme, regardless of the nature of the donor, those interactions where the $\sigma^*_{\text{O-H}}$ act as acceptors of charge (interactions 1, 3, 5) are well separated from the rest, having the smallest X...H distances, the largest accumulations of electron densities, the largest virial

ratios, and the largest $-E_{d \rightarrow a}^{(2)}$ energies, as well as being the only ones with negative bond degree parameters ($\mathcal{H}/\rho$) and consistently showing stronger character than the archetypal $\text{O}\cdots\text{H}-\text{O}$ hydrogen bond in the water dimer, in which the acceptor is also a $\sigma_{\text{O}-\text{H}}^*$ orbital and whose associated values are included in all plots as reference. However, not all primary hydrogen bonds are equal: when $\sigma_{\text{O}-\text{H}}^*$ orbitals are involved (interactions 2, 4, 6), the associated interactions energies are smaller, to the point that all quantitative descriptors of bonding associated to these orbitals fall in the same ranges as the weaker interactions. In the same line, because of their relative abilities to transfer charge to the $\sigma_{\text{H}-\text{O}}^*$, $\sigma_{\text{H}-\text{N}}^*$ orbitals, we introduce a $\text{N} > \text{C}=\text{O} > \text{O}-\text{H}$ "bondphile" scale in primary hydrogen bonds. Under these circumstances, a general stabilization principle in the Alanine dimers, with the most significant component being the transfer of minute amounts of charge from the amino group towards the $\text{O}-\text{H}$ bond, may be established.

2.2.2. Secondary Hydrogen Bonds

Secondary hydrogen bonds are listed as interactions 7–12 in Table 1 and in Figure 2. These are weak interactions, characterized by quantitative descriptors that place them in close proximity or below the strength of the reference water dimer (Figure 3). There is a large number of secondary hydrogen bonds, in fact, they are the most common type of interaction found in the Alanine dimers, with, *a priori*, three times as many $\text{H}-\text{C}_\beta$ possibilities as $\text{H}-\text{C}_\alpha$, $\text{H}-\text{C}_\alpha$ and $\text{H}-\text{C}_\beta$ bonds seem equally susceptible to accept small amounts of charge by increasing the occupation of the corresponding antibonding σ^* orbitals. One very important point is that despite the relative weak nature of secondary hydrogen bonds, their cumulative role in the stabilization of Alanine dimers should not be ignored. This is a very sensitive point that translates into larger systems in the form of large fluxional walls of stabilizing non-Covalent interactions. Fluxional walls of charge, which are clearly attractive in our case (see the green surfaces in Figure 4) may be defined as the non-rigid (hence fluxional in analogy to fluxional molecules) surfaces arising from charge displacement to the interstitial region between fragments. These fluxional walls are pivotal for example in the initial molecular recognition and attachment of SARS-COV-2 to the ACE2 receptors in the cell membranes of eukaryotic cells,^[6] in the insertion of non-steroidal drugs into cell membranes,^[60,61] and heavily contribute to the molecular basis for hydrophobicity.^[62] For the case of Alanine dimers, the fluxional walls of non-Covalent interactions which collectively include primary and secondary hydrogen bonds, dihydrogen contacts, and non conventional or exotic interactions, are shown in Figure 4 for the chosen set of 32 dimers and for the three dimers with the largest numbers of intermolecular contacts. It is clear from this representations that the cumulative effect of a number of very weak individual interactions, each transferring a tiny amount of electron density into the interstitial region separating the monomers, leads to this fluxional wall of non-Covalent attractive interactions, in

most cases completely responsible for the stabilization of the dimers.

2.2.3. Dihydrogen Contacts

Interactions 13–19 in Table 1 and in Figure 2. Dihydrogen bonds are very weak intermolecular interactions whose cumulative effect is responsible for quite strong stabilizations. This counterintuitive effect is very well documented and supported by experimental evidence for example in the increasing binding energies as the number of carbon atoms increases in the methane to decane series,^[63,64] which leads to a 98.4 °C boiling point for heptane at room conditions and in the large cohesion energies for ammonia borane.^[65,66] Only intermolecular dihydrogen contacts are possible for both substances. In the case of Alanine dimers, Figure 3 shows that interatomic distances for dihydrogen bonds are comparable to secondary hydrogen bonds, but dihydrogen bonds are considerably weaker, having the smallest densities and Laplacians at bond critical points (tallest peaks at the left in both cases) and virial ratios (leftmost peak). In fact, NBO energies are so small that an inset in the corresponding plot is necessary. In the NBO donor→acceptor scheme, dihydrogen bonds are always described by charge transfer between the bonding and antibonding orbitals in the involved σ bonds, that is, charge transfer in dihydrogen bonds is a two way process that for a given pair of $\text{X}-\text{H}$, $\text{Y}-\text{H}$ bonds satisfies $\sigma_{\text{X}-\text{H}} \rightarrow \sigma_{\text{H}-\text{Y}}^*$ and $\sigma_{\text{Y}-\text{H}} \rightarrow \sigma_{\text{H}-\text{X}}^*$ (notice that Table 1 only lists a few representative cases thus some of the orbital interactions for dihydrogen bonds are not included). Although dihydrogen bonds are somewhat exotic for the non specialists, a good deal of scientific literature has been devoted to understand their properties.^[44,45]

2.2.4. Non Conventional or Exotic Contacts

We find it very interesting that well defined bonding paths and associated orbital interactions connecting heavy atoms (interactions 20–31) and also $\text{C}\cdots\text{H}$ atoms (interaction 32) from the two monomers are consistently found. The large number of occurrences (216 cases or $\approx 14\%$ of the total) and the corresponding interaction energies, of the same magnitude as secondary hydrogen bonds and dihydrogen contacts, indicate that those interactions are not to be taken as mere curiosities and that they indeed play a role in the stabilization of Alanine dimers. These interactions are seldom mentioned in the scientific literature,^[67–71] and thus we analyze them in some detail next.

As listed in Table 1, we found six types of non conventional intermolecular contacts and collect their bonding descriptors into a single set in the plots of Figure 3. Interaction distances for non conventional contacts are quite large (the rightmost peak in the left top panel of Figure 3), centered at $\approx 3 \text{ \AA}$ and extending beyond $3.5 \text{ \AA}$. However, because these are atoms with multiple electrons, when compared to interactions involving hydrogen atoms (i.e. hydrogen bonds and dihydrogen

contacts), the associated bond critical points accumulate large amounts of electron densities for such long distances, nonetheless, the virial ratios, bond degree parameters and NBO energies still categorize this collective set of interactions as of weak, long range character.

The classical description of hydrogen bonds invokes an electrostatic $Y^{\delta-} \cdots H^{\delta+}-X^{\delta-}$ interaction which would become a destabilizing $Y^{\delta-} \cdots X^{\delta-}-H^{\delta+}$ factor in the non conventional cases. Weinhold and Klein^[72] reported for the first time anti-electrostatic hydrogen bonds and showed that orbital interactions more than compensate for the local Coulomb repulsion, other studies soon followed.^[73–76] For the particular case of the Alanine dimers, in the $\phi_d \rightarrow \phi_a$ NBO orbital interaction picture, it is the $\pi^*_{C=O}$ orbital that acts as the acceptor of charge for most of the non conventional contacts. However, when one of the bonds involved has a proton (C–H, N–H, O–H, interactions **26**, **28**, **30**, **31**) the situation is quite interesting as illustrated in Figure 5: we argue that despite no proton being sandwiched in between the two electronegative atoms in the anti-electrostatic case (right panel), this is still a hydrogen bond because the exact same two orbitals are involved in both cases with a subtle difference: in the direct $n_Y \rightarrow \sigma^*_{H-X}$ case (left panel) the two orbitals overlap using the H–end of the antibonding orbital and in what we call the “inverted” $n_Y \rightarrow \sigma^*_{H-X}$ hydrogen bond (the anti-electrostatic case), the same two orbitals overlap using the X–end of the acceptor orbital. Notice that the argument still holds for interaction **30** in which a σ^*_{H-N} orbital acts as the donor of charge.

3. Summary and Conclusions

Alanine is the second smallest among the set of 20 amino acids constituent of all proteins, nonetheless, a stunning structural diversity for the Alanine dimers is uncovered in this work after exhaustive stochastic explorations of a statistically representative sample of 10 out of the possible 351 binary combinations of the 13 reported isolated monomers. A total of 416 well defined equilibrium structures were found. This large set of dimers is tightly packed within a $\Delta\Delta G \approx 16 \text{ kcal mol}^{-1}$ window relative to the triply degenerate putative global minima. The

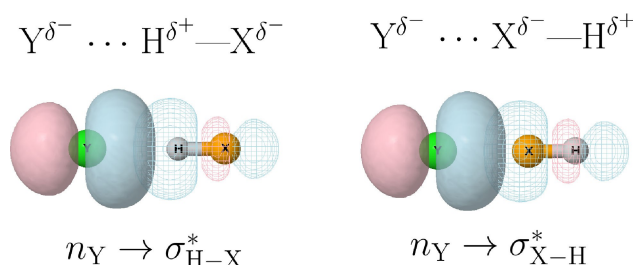


Figure 5. Direct $Y \cdots H-X$ hydrogen bond (left) and its counterpart anti-electrostatic $Y \cdots X-H$ exotic contact (right). Despite the fact that no hydrogen atom is sandwiched in between the two electronegative atoms, from the point of view of orbital interactions, the anti-electrostatic contact is actually an “inverted” hydrogen bond with the n and σ^* orbitals overlapping from opposite ends as in the direct case.

dimers are strongly bound, with binding energies of up to $13.4 \text{ kcal mol}^{-1}$ at the DLPNO-CCSD(T)/6–311++G(d,p) level corrected by the basis set superposition error and by the zero point vibrational energies over B3LYP–D3 geometries. A complex picture of monomer to monomer bonding emerges because a set of 32 different types of intermolecular interactions are at play stabilizing the dimers, these interactions may collectively be classified as primary $Y \cdots H-X$ and secondary $Y \cdots H-C$ hydrogen bonds, $H \cdots H$ dihydrogen contacts, and non conventional or exotic $X^{\delta-} \cdots Y^{\delta-}$ interactions. These exotic, anti-electrostatic interactions are not uncommon, constituting $\approx 14\%$ of the total number of intermolecular contacts. Non conventional contacts are mediated by orbital interactions that overcompensate local Coulomb repulsions in an otherwise destabilizing situation. All descriptors of the nature and strength of intermolecular bonding interactions place them in the weak category, except for a few cases of primary hydrogen bonds, nonetheless, their cumulative action results in the large stabilization energies. The mechanism of this stabilization is the transfer of tiny amounts of electron charge from the monomers to the interstitial region, which results in large fluxional walls of attractive non-Covalent interactions.

Final remark. We argue that our findings are deeply connected with the fundamental problems of the origin of life and its evolution, which has lead to its present diversity and abundance, via the second law of thermodynamics (see the formal treatment by England of the thermodynamics of self-replication^[77]). The point is that the local reduction in entropy needed for the emergence, support, and replication of highly organized living organisms is overcompensated by a larger entropy increase of the universe. This increase of the entropy of the universe is due to the overwhelmingly large number of realizable microstates and possibilities in all aspects of the problem treated here: from the large number of protein primary structures (recall 2.4×10^{18} possible primary sequences involving just 20 aminoacids, each with its own secondary and tertiary structure complexity), to the almost unthinkable number of protein–protein interactions, each having an astonishing large number of individual contacts as can be inferred from the 1578 intermolecular contacts spread among 32 interaction types on the 416 dimers of modest Alanine. We can go even deeper in this entropy increasing regression by invoking the large number of possibilities for donor→acceptor orbital interactions not even mentioned in this work (we only showed the largest ones) or that may result from larger basis sets within the NBO paradigm. If pressed to the limit, we may reach the final frontier of the very foundations of quantum mechanics and think that this abundance of possibilities is perfectly consistent with the superposition principle and with the probabilistic interpretation of the measurement postulates. Finally, it should be noticed that the large number of possibilities exceedingly favors molecular evolution towards systems of increasing abilities to survive local pressures because even if specific mutation probabilities are small, they will eventually occur in this large sea of possibilities and will eventually become the dominant traits.

Supporting Information Available

Binding energies and Cartesian coordinates for the entire set of 416 dimers are provided. A relaxed scan and accurate energies for the monomers are also provided to justify the choice of model chemistry.

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Conflict of Interest

The authors declare no conflict of interest.

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